

Module 1 - Q2: MLflow Experiment Comparison

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MLflow Run Comparison — experiment: mnist-mlp (6 runs)

run	hidden	lr_init	alpha	max_iter	accuracy	f1_macro	train_acc	train_loss
mlp-h(128,)-lr0.01	(128,)	0.01	0.0001	20	0.9425	0.9408	1.0000	0.0014
mlp-h(256,)-lr0.001	(256,)	0.001	0.0001	20	0.9383	0.9366	0.9981	0.0332
mlp-h(64,)-lr0.01	(64,)	0.01	0.0001	20	0.9367	0.9344	1.0000	0.0024
mlp-h(128, 64)-lr0.001	(128, 64)	0.001	0.0001	20	0.9317	0.9299	0.9977	0.0252
mlp-h(128,)-lr0.001	(128,)	0.001	0.0001	20	0.9267	0.9249	0.9894	0.0687
mlp-h(64,)-lr0.001	(64,)	0.001	0.0001	20	0.9225	0.9204	0.9765	0.1164

Analysis

The best-performing run was run 5 (a single hidden layer of 128 units with `learning_rate_init` = 0.01), reaching a test accuracy of 0.9425 and macro-F1 of 0.9408. Its advantage came almost entirely from the higher learning rate: within the fixed 20-iteration budget, `lr` = 0.01 converged substantially further than 0.001, so the network reached a better solution before training stopped. Depth did not help — the two-layer (128, 64) network (run 6) scored only 0.9317, below run 5, indicating the extra layer added capacity without benefit at this data and iteration budget.

Overfitting is clearly visible. For both `lr` = 0.01 runs (4 and 5) the training accuracy reached 1.0000 while test accuracy stayed near 0.94 — a gap of roughly six percentage points — and the logged `train_loss` fell close to zero. The model memorises the 4,800-sample training set faster than it generalises, and the gap widens as learning rate and capacity grow.

Of the two swept hyperparameters, `learning_rate_init` had the larger and more consistent effect: raising it from 0.001 to 0.01 improved test accuracy for both the 64-unit (0.9225 → 0.9367) and 128-unit (0.9267 → 0.9425) networks. Hidden-layer size helped at the low learning rate (64 → 256 units: 0.9225 → 0.9383) but its effect was smaller and non-monotonic. Learning rate therefore dominated performance.

Logging code added to the script

```
mlflow.log_param("hidden_layer_sizes", hidden_layer_sizes)
mlflow.log_param("learning_rate_init", learning_rate_init)
mlflow.log_param("alpha", alpha)
mlflow.log_param("max_iter", max_iter)

mlflow.log_metric("accuracy", acc)
mlflow.log_metric("f1_macro", f1)
mlflow.log_metric("train_accuracy", train_acc)
```

```
mlflow.log_metric("train_loss", model.loss_)
```